Point-to-Point Protocol

(PPP)

What is PPP?

- PPP stands for Point-to-Point Protocol.
- It is a data link layer protocol used to establish a direct connection between two network nodes.
- Commonly used for dial-up, DSL, and VPN connections.

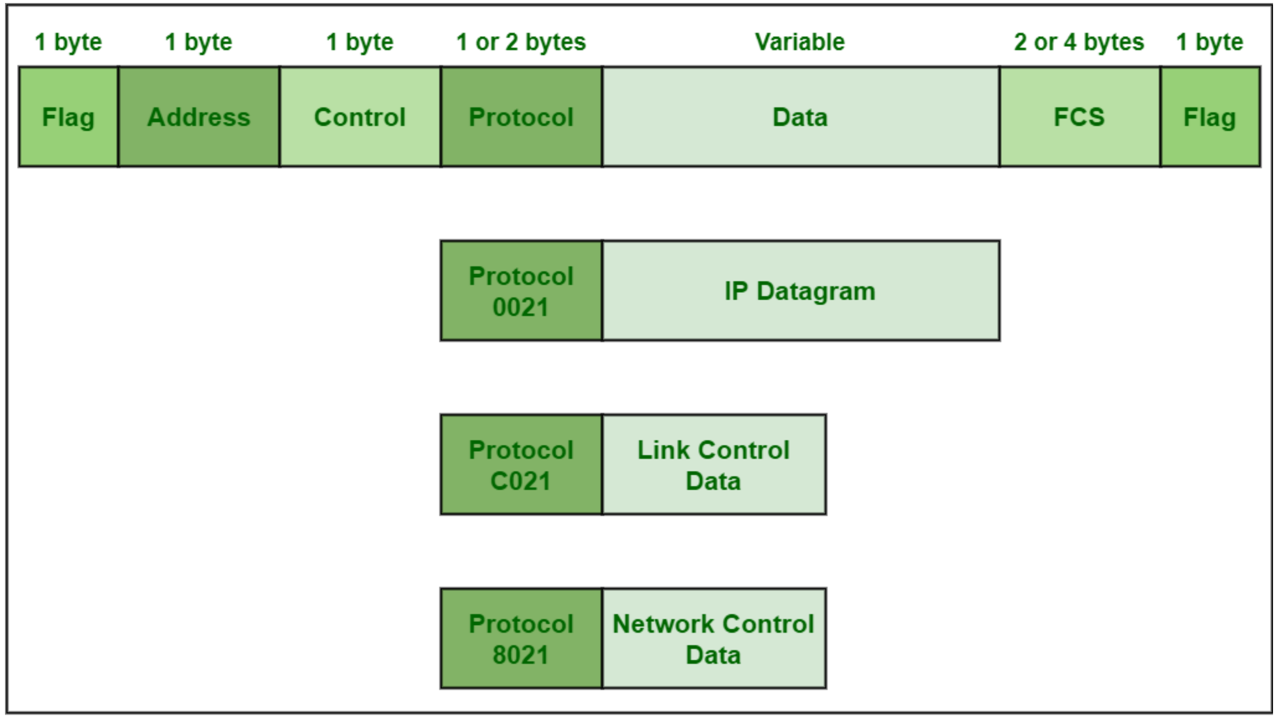
Purpose of PPP

- To transmit packets over serial links or point-to-point connections
- To encapsulate multiple network layer protocols (like IP, IPX, AppleTalk)
- To provide authentication, error detection, and link management

Features of PPP

- Works over synchronous and asynchronous links
- Supports multiple network protocols using NCP
- Provides error detection using CRC
- Supports authentication (PAP or CHAP)
- Simple and widely used for point-to-point communication

PPP Frame Structure



PPP Frame Format

PPP Frame consists of:

Field	Purpose
Flag	Marks the start/end of the frame (01111110)
Address	Usually set to 11111111 -all 1's simply indicates that all of the stations are ready to accept frame
Control	Usually 00000011 (unnumbered frame) -This 1 byte is used for a connection-less data link.
Protocol	Identifies the network protocol (IP, IPX, etc.)
Data	Encapsulated payload (network layer packet)
FCS (Frame Check Sequence)	Error detection

Components of PPP

1. LCP (Link Control Protocol)

- Establishes, configures, tests, and terminates the link
- Handles authentication and link options

2. NCP (Network Control Protocol)

- Configures network layer protocols (IP, IPX, AppleTalk)
- Allows multiple protocols over the same link

Authentication in PPP

1. PAP (Password Authentication Protocol)

- Simple, uses username and password
- Password is sent in clear text

2. CHAP (Challenge Handshake Authentication Protocol)

- More secure
- Uses challenge-response mechanism
- Password is not sent directly

Advantages of PPP

- Supports multiple protocols
- Provides authentication and error detection
- Can work over different physical media
- Simple and widely supported

Summary

- PPP = Point-to-Point Protocol
- Works at data link layer to connect two nodes
- Uses LCP for link setup and NCP for network protocol setup
- Can use PAP or CHAP for authentication
- Widely used in dial-up, DSL, and VPN connections